

ATHENA

8-BIT COMPUTER PROJECT

Engineering Record & Learning Journal

DESIGN • BUILD • VERIFY • OBSERVE • UNDERSTAND

A practical, evidence-led computer engineering project exploring the design and construction of an observable 8-bit computer system from first principles.

Clock • CPU • Single-T-State Stepping • Buses • Memory • I/O • Firmware • Measurements • Faults •
Design Decisions • Schematics • Test Evidence

Project owner Edward Croasdell

Started 2026

Repository [efcroasdell/Athena-8-Bit-Computer](#)

Record Printed master binder supported by the online
engineering repository

UNDERSTANDING BEFORE COMPLETION

*The project is successful when each subsystem can be explained, predicted, measured,
diagnosed and deliberately modified.*

ATHENA

DOCUMENT CHANGelog

Printed Ring-Binder Record

This page records substantive changes to the printed Athena project documentation. The GitHub repository remains the authoritative online engineering record; the binder is a maintained physical project record.

Date	Revision	Documents	Change
18 Sep 2026	1.1	Front matter; Clock Module record; Z80 CPU Bring-up record	Initial binder cover and technical records established. Documentation updated to state explicitly that MANUAL/STEP produces one CPU_CLK pulse per press and therefore permits a connected Z80 to advance one T-state at a time for timing-diagram, bus and control-signal observation.

Current printed set

- Front cover and changelog
- Athena Clock Module — Build, Verification and Learning Record
- Athena Z80 CPU Bring-up — Progress, Evidence and Learning Record

Change-control principle

Corrections, new measurements and later as-built changes should be dated and recorded without erasing the historical sequence that led to them.